

Algebra 1: Unit 9, Lesson 1

1. A quadratic function has the equation, $y = a(x + 3)(x - 5)$ and passes through the point, (6,45). What is the value of a ?

2. Gerardo found the partially completed equation of a quadratic function in the form $y = -\frac{1}{3}(x - p)(x - q)$ with the x -intercepts of (3,0) and (-6,0).

What is the equation of the quadratic function in standard form?

3. Gerardo is writing the equation for the parabola whose x -intercepts are (1,0) and (-5,0) that passes through the point (3, -24). He uses the equation, $y = a(x - p)(x - q)$. Gerardo tells his partner that he knows that he can use, $p = 1$, $q = -5$, $x = 3$ and $y = -24$, to find the value of a .

What is the value of a ?